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## Enabling real-time adaptations for individualized customer experience in user-centered e-business applications

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### Abstract

Optimized user experience and individualized task assistance are key factors for the success of e-business applications because they serve both, users, and service providers. In this paper we present an adaptation agent for observing and analyzing interactive behavior, emotional state and task-accomplishment of individual users while working with interactive e-business applications. The agent recognizes specific situations and can adapt the interactive application at runtime by co-operating with a model-based user interface development framework (MBUID) and by providing domain-specific guidance for the user. After introducing purpose, business context of our approach, and related work, the paper discusses the structure and functionality of the SitAdapt adaptation agent. The operation of the adaptation agent is demonstrated for a travel-booking platform that currently serves as a workbench for testing and evaluating our situation-analytic and emotion-triggered approach for creating highly competitive and individualized business applications.

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## 1. Introduction

Optimized user experience during the whole online customer journey is a key factor for successful e-business applications with the focus on the individual user and her or his interactions with the involved business processes. By observing the user before, during, and after interactive sessions with the e-business system and by recognizing typical situations and dynamic changes of user needs, an intelligent adaptation agent can continuously analyze the user's actions, interactive behavior, and emotional state. By matching actual user interactions and accessed user interface and business objects with the potential goals and sub-tasks that were modeled in the task model of the e-business application the agent can recognize the current state of task accomplishment and/or find out, why a goal was not reached successfully.

Tracking the user anonymously during his or her web sessions might be helpful for getting statistical usage data and leading users to specific content, but cannot capture the full individual behavior, cognitive load, and current mood of a specific user. The SitAdapt adaptation agent architecture [1] presented in this paper is a user- and situation-aware lab environment for individualized real-time adaptations of interactive systems. The system has been steadily improved over a series of evolutionary steps and was evaluated with application prototypes from different target domains. The current version of the SitAdapt agent has been optimized for e-business web applications and provides support for real-time adaptation of generic and domain-specific parts of such applications. We use the SitAdapt agent architecture to exploit the potential of individualized online customer journeys by applying it to existing commercial e-business websites [2] and to our own experimental e-commerce workbench applications. SitAdapt distinguishes between the following temporal contexts:

- Before the session the agent can use observational data to tune the system in to the current mood of the user and adapt the user interface to the specific needs of the user, e.g., age-related aspects like font-size, layout, or language style as well as mood-related styles and aesthetics.
- At runtime the agent can decide to adapt the user interface in real-time to improve user experience or give guidance for the user's task management [3]. For this purpose and to arrive at reliable decisions SitAdapt evaluates the inputs of different sensors and information about the current application context at the same time.
- After the session the agent can draw conclusions from the observed interactive behavior and emotional patterns to improve the user interface or even the task structure of the system before the next session of the same user.

Also, information about different users can be gathered and statistically evaluated to discover frequent generic and domain-specific behavioral patterns when using the same web application. This can lead to the automatic discovery of typical user personas. Such data can also be analyzed to find the reasons why real users sometimes don't follow the task and sub-task sequences as proposed by the task model of the application. Therefore, an adaptation approach for e-business applications with large interactive parts offers numerous benefits both for individual users and service providers.

### 1.1. Related Work

In general, three different categories of user interface adaptations can be distinguished [4], [5]:

- Adaptable user interfaces. The user customizes the user interface to his or her personal preferences.
- Semi-automated adaptive user interfaces. The user interface provides recommendations for adaptations. The user must decide whether he or she wants to accept the recommendation or not.
- Automated adaptive user interfaces. The user interface automatically reacts to changes in the context-of-use.

Adapt-UI [6], like SitAdapt, offers runtime-adaptation of the user interface. The system focuses on context-of-use adaptations when migrating to other devices and platforms, but also manages some user-related aspects. The system includes models of the user interface, the context, and the possible adaptations. Context-changes are triggered with adaptation rules. Such rules can show or hide pre-modeled UI elements, change navigation paths in pre-modeled ways, and react to a few simple user-related aspects that are implemented by accessing a face detection library.

Adaptation rules are also used in [7]. In this approach an adaptation framework was developed that is based on a detailed problem space model focusing on the context of use and the temporal aspects of emotions that trigger the adaptations of websites. An adaptation architecture that performs adaptation cycles on the website was designed. It consists of components for watching the user, administrating the context of use (user, platform, environment) and possible UI adaptations, executing adaptation rules, and finally modifying the user interface at the end of an adaptation cycle if necessary. The architecture was tested with a serious game application. The underlying problem space model distinguishes between:

- different emotion models, e.g., Ekman basic emotions,
- different emotion sources, e.g., automatically detected peak facial emotion  $x$ ,
- different temporal inference categories, e.g., inferred from values in the last  $y$  seconds, and
- different temporal application types, e.g., applied if a duration of  $z$  seconds was observed.

Adaptation rules can be defined using the possible attributes of the problem space classes and can also access the sensors within the adaptation architecture.

By adding the concept of situations that include and exploit the variability of the cognitive and emotional states of the user to the adaptation modeling process, a new category of individual adaptations is made possible. With SitAdapt situation-aware adaptations can be modeled by the developer and generated or activated at runtime.

To capture the individual requirements of a situation, [8] has suggested that a situation specification must cover the user's operational environment  $E$ , the user's social behavior  $B$ , by interpreting his or her actions, and a hidden context  $M$  that includes the user's mental states and emotions. A situation  $Sit$  at a given time  $t$  can thus be defined as  $Sit = \langle M, B, E \rangle_t$ . A user's intention for using a specific software service for reaching a goal can then be formulated as a temporal sequence  $\langle Sit_1, Sit_2, Sit_3 \rangle$ , where  $Sit_1$  is the situation that triggers the usage of a service or a task execution and  $Sit_n$  is the goal-satisfying situation.

This model provides a good starting point for the construction of situation-aware applications. There are two different, but strongly related views on situation-awareness. Situation-awareness as discussed in [9] and many other approaches, supports the user to correctly interpret her current situation by letting her focus on the most important contextual and situation-typical information, e.g., in complex and safety-critical human-machine interactions. The other view, as taken by SitAdapt, is a situation-analytic view, where the system is observing the user, the interactive software, and the environmental and platform-related context. The system recognizes specific situations and combinations of situations and adapts the system in real-time to optimize task accomplishment and user experience.

However, for recognizing and evaluating situations, the hidden context  $M$ , i.e., the user's emotions, intentions and cognitive load must be made visible and exploitable. Adaptive situation-aware software systems need cognitive and analytic capabilities to interpret the multitude of available emotional and bio-physical signals as well as application and contextual meta-data, all of them changing over time, to being able to infer a goal-reaching set of adaptations. SitAdapt chooses a combined approach for interpreting the hidden context:

- Fine grained visual, eye-tracking, bio-physical, and emotion monitoring of the interactive user session together with application meta-data and raw data storage
- Extracting coarse grained situation profiles from the raw data with time-stamped event and event duration data. Linking the application meta-data in situation profiles to the relevant models of the interactive application.
- Selecting situation rules if typical events or significant attribute value changes arise
- Moderation of conflicting rules and situation rule firing leading to the generation or activation of adaptations of the interactive system.

## 1.2. Methods

To assess the quality of the overall adaptation process and the domain-specific adaptations, for each e-commerce application domain under research a web application prototype is constructed, or an existing web application is instrumented to provide user interface adaptivity. Application prototypes are developed using an iterative agile process that is combined with user-centered design activities [10]. Usability engineering and UX experts accompany each

iteration of the process. When the new application prototype or the instrumented application becomes available, user studies are carried out that include questionnaires and scenario-based usability and UX tests. The tests can directly access the observation data of the SitAdapt agent. The results of these studies typically lead to improved adaptation capabilities, better adaptation rules, and optimized user interfaces of the examined e-commerce application prototypes. Also, the architecture of the SitAdapt agent itself has been modified several times after such studies. Examples of two experimental user studies where SitAdapt was used in a real-world e-commerce platform for cosmetics products are discussed in [2]. Currently, a similar evaluation study is under way for the travel-booking application discussed in section 2. The results of this study will be presented in a future paper.

### 1.3. SitAdapt Agent Structure and Functionality

SitAdapt [1], [11] is an architecture and runtime-system for adaptation agents that are running together with all kinds of interactive applications. The agent observes the user and his or her interactions with the application using various visual and bio-physical devices. The raw data frames are recorded with a resolution of 1/60 s. The essential user behavior and temporal changes are extracted from the raw data frames over time and stored in situation profiles in a database. Situation profiles also hold links to the user model, the abstract, concrete, and final user models, the task model, the environment model, and other models involved in the current situation. A situation profile for a ticket booking application, e.g., can have the following (simplified) structure:

```
<SituationProfile>
  <TargetApplication>
  <User>
  <Situation_booking>
    <SituationTime> 30 s
    <FUI_link> Wizard_Part_1
    <CUI_link> model_CUI_1
    <AUI_link> model_AUI_1
    <Dialog_link> order_ticket
    <Task_link> order_ticket
    <Concept_link> concept_1
    <Eye_Tracking> ...
    <EmotionalState> ...
    <UserPsychologicalState> tired
    <BiometricState>
      <Pulse> normal
      <StressLevel> orange
  </Situation_booking>
```

Situation rules are triggered when typical situations, i.e., significant changes of observational data values or combinations of such values over a specified time span, are met by a situation profile at runtime. Also, specific events, e.g., eye-tracking coverage of a specific user interface object for a specified time span, can lead to the selection of a situation rule. If more than one rules apply, SitAdapt decides which of the rules should be fired and generates the needed modifications of the UI and/or the active sub-task model. In the above example, a situation rule could be found that recognizes that the user is tired and activates a less complex input form.

The resources for modeling and generating real-time adaptations, e.g., domain models (task and concept, i.e., business class model), user interface models (abstract, concrete, and final user interface model), context-of-use models (user, device, environment, UI toolkit, and design model) as well as generic and domain-specific user interface patterns and pre-modeled adaptation templates are provided by the PaMGIS MBUID framework [12]. Fig. 1 shows the architecture of the SitAdapt agent with all monitoring hard- and software-sensors and its interfaces to PaMGIS, needed when creating rules and when generating the adapted user interface.

SitAdapt consists of the following components [11]:

- The *data interfaces* use the different APIs of the devices (eye-tracker, wristband, facial expression recognition software interface, metadata from the application) to collect data about the user. Two different data types for generation and adaptation of the user interface received from the different input devices are supported: atomic data types as constant attribute values (e.g., *age* = 40, or *glasses* = *true*) and temporal data types. Temporal data management makes it possible to document and analyze changes in the recorded data by time-stamping these data. This allows to determine which value was valid at what time and how long a value range was held. For instance, *emotion type*, *blood pressure* or *eye positions*. Atomic and temporal data can be used to create rules, that influence the adaptation of the user interface.
- The *recording component* combines and synchronizes the different input records with a timestamp.

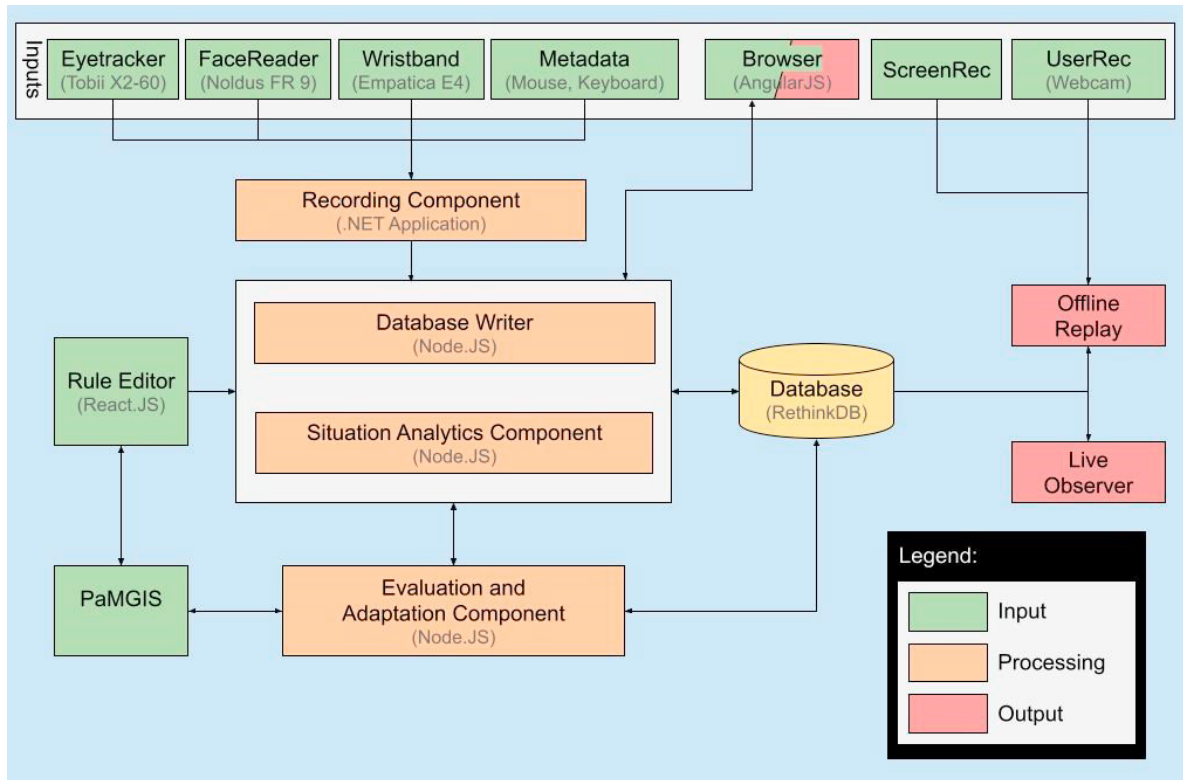


Fig. 1. SitAdapt Agent Structure with interfaces.

- The *database writer* stores the data from the recording component and from the browser in the database, where the raw situations and situation profiles are managed. It also controls the communication with the rule editor.
- The *rule editor* allows the definition and modification of situation rules, e.g., for specifying the different user states and the resulting actions. The rule editor is very flexible and can use all input data types and attribute values as well as their temporal changes for formulating rule conditions. At runtime rules are selected by the situation analytics component for adapting the user interface if the conditions of one or more rules apply. Situation rules can also activate HCI-patterns in the PaMGIS pattern repository. These patterns come with different levels of abstraction. They may contain concrete templates for generating interaction objects or information that describes, how low-level user interface attributes can be modified.
- The *situation analytics component* analyzes and assesses situations by exploiting the observed data. Situation rules are selected by the situation analytics component when the rule conditions are satisfied. Situation rules interact with the situation profiles stored in the adaptation agent's database. The rule actions either directly trigger simple adaptations or interact with the PaMGIS resources as described above.

- The evaluation and decision component uses the data that are provided by the situation analytics component to decide whether an adaptation of the user interface is currently meaningful and necessary. For this purpose, the component evaluates one or more applicable situation rules and must solve possible conflicts between the rules. Whether an adaptation is meaningful depends on the predefined purpose of the situation-aware target application. Such goals can be detected, if one or more situations in the situation profile trigger an application-dependent or domain-independent situation rule. Situation rules are related to patterns. They define behavioral and context-related situational patterns. If the decision component decides that a complex adaptation is necessary, it must provide the artifacts from the PaMGIS pattern and model repositories to allow for the modification of the target application by the adaptation component.
- The *adaptation component* generates the necessary modifications of the interactive target user application.

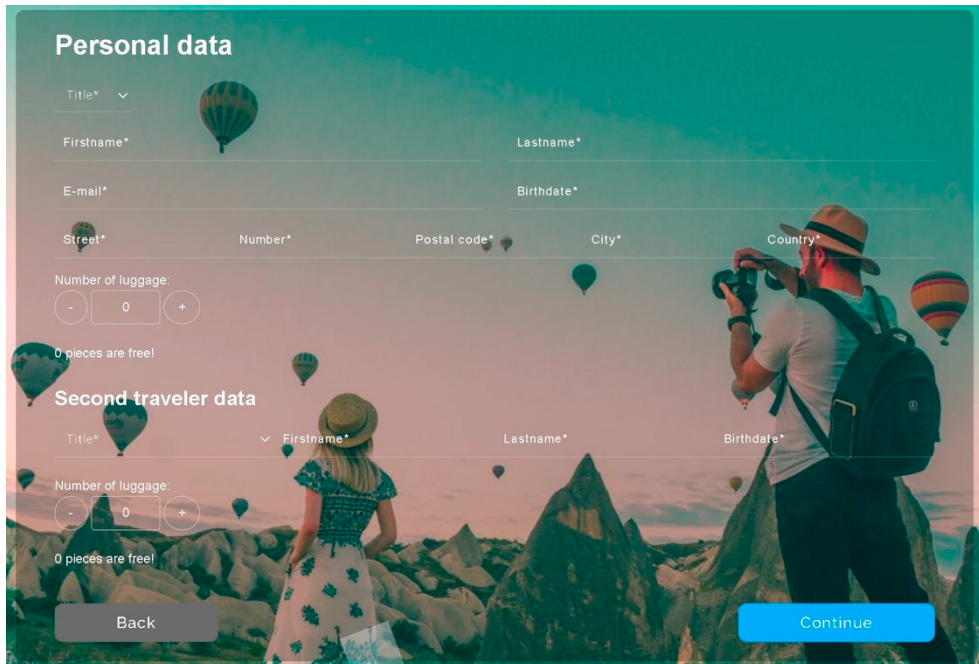
## 2. The SitAdapt Agent at Work

By accessing the tools and modeling resources of the PaMGIS MBUID framework, the developer can model different types of adaptations that are generated at runtime, if a situation arises, where a situation rule is fired. The fired rule activates the generation of a specific adaptation in the final user interface. The SitAdapt agent allows different types of automated user interface adaptations.

- *Changes of media and interaction object types.* Such changes modify the way of the information presentation. Different types of charts (mosaic or a bar charts), personalized areas or functions (areas with functions frequently used by the individual) can be chosen by situation-aware adaptations. Another example for an adaptation is the change of the input medium from text to speech.
- *Changes in content.* The agent can adapt the content presented to the user. Thus, it is possible to change button labels or offer help texts. In the e-commerce domain, specific advertising and offers, as well as additional elements or pop-ups such as vouchers can be presented to the user. Different users can also be offered different content, e.g., variably detailed text information or product images, or variably detailed content presentations, by adding or removing content attributes.
- *Changes in design and layout.* The agent can adapt the design of the program or website, e.g., colors, background colors, images and the contrast or change text layout and font size and type. The system can also change the layout of the user interface objects.
- *Changes in structure.* SitAdapt can change the number of objects displayed on the interface, for example the number of input fields of a form.
- *Personalized information.* The system can change the content of the text. For example, more formal or easier language.

There is also the possibility of combining the various types of adaptations. In the following an example for an adaptation of the category *changes of media and interaction object types* is given. The shown travel-booking portal is implemented in Python and with the Django framework. A JavaScript function is responsible for interface modifications. The graphics or structural changes are already known to the web application because they were modeled and coded in the development phase. They are made visible to the user at runtime by changing a variable in the HTML file after the travel portal has received the information from the rule editor to show the wizard via the WebSocket server. The decision to trigger a specific rule is made, because the SitAdapt agent has observed the user in a specific situation of interacting with the system, where the rule is applicable.

Here, the SitAdapt agent might have recognized a platform change from laptop to mobile phone, or, by consulting the user model, that a user with no booking experience is working with the travel-booking platform. In this case a situation rule is fired that changes the user interface for personal data input (Fig. 2) into a wizard-based input object (Fig. 3).



The image shows a 'Personal data' form overlaid on a scenic background of a person taking a photo of a landscape with hot air balloons. The form is divided into two sections: 'Personal data' and 'Second traveler data'. Each section contains fields for Title\*, Firstname\*, Lastname\*, Birthdate\*, E-mail\*, Street\*, Number\*, Postal code\*, City\*, and Country\*. There are also 'Number of luggage' input fields with minus, zero, and plus buttons, and a message '0 pieces are free!'. At the bottom of each section are 'Back' and 'Continue' buttons.

**Personal data**

Title\*

Firstname\*

Lastname\*

Birthdate\*

E-mail\*

Street\*

Number\*

Postal code\*

City\*

Country\*

Number of luggage:

0 pieces are free!

**Second traveler data**

Title\*

Firstname\*

Lastname\*

Birthdate\*

E-mail\*

Street\*

Number\*

Postal code\*

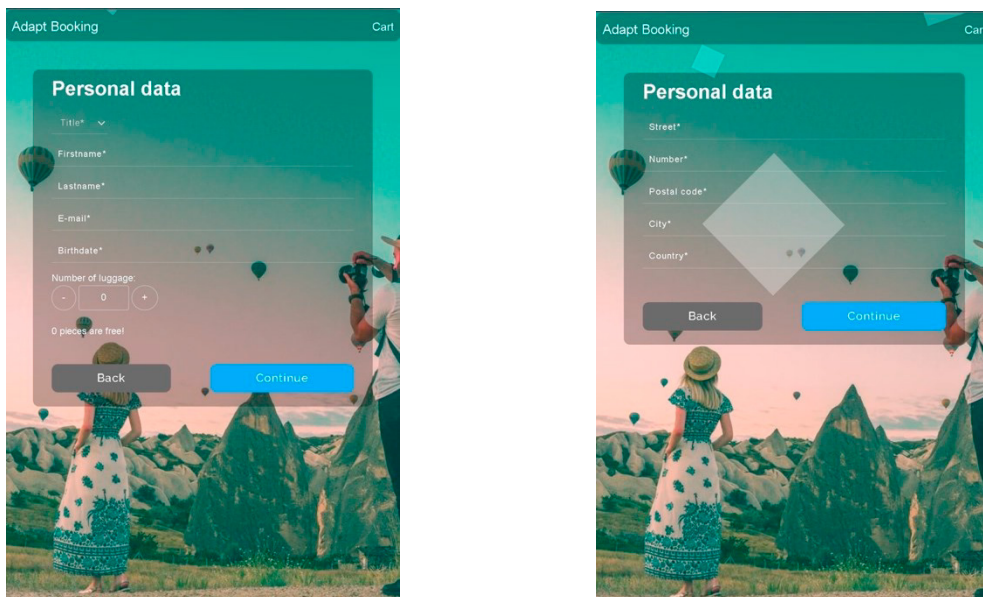
City\*

Country\*

Number of luggage:

0 pieces are free!

Fig. 2. Personal data input before adaptation.



The image shows two side-by-side screenshots of the 'Personal data' form after real-time adaptation. The left screenshot shows the full form with all fields visible. The right screenshot shows the form with a large, semi-transparent diamond shape overlaid on the 'Number' and 'Postal code' fields, indicating a focus or selection. The 'Back' and 'Continue' buttons are visible at the bottom of the form.

**Personal data**

Title\*

Firstname\*

Lastname\*

Birthdate\*

E-mail\*

Street\*

Number\*

Postal code\*

City\*

Country\*

Number of luggage:

0 pieces are free!

Fig. 3. Wizard-based input after real-time adaptation.

### 3. Conclusion and Future Work

In this paper we have presented the SitAdapt adaptation agent in its current version as a lab-based environment for individualizing interactive e-business applications. We have implemented adaptation processes for several e-commerce domains, e.g., web-shops, ticket-booking, and travel-booking applications. With the SitAdapt approach we have successfully strived for our two main goals at the same time:

1. Find new architectures and create new frameworks for individualized real-time adaptation processes for interactive systems, and,
2. Building an environment for studying the most promising ways to optimize user experience and individualize the online customer journey for large e-business applications.

We are currently focusing our research on combining visual and EEG-based emotion recognition to receive even more reliable data about user emotions. We recently have developed a new approach for high-quality real-time emotion recognition that requires only seven EEG channels and little computing power [13]. We are planning to implement this approach with a standard headset that provides the necessary sensor electrodes and to integrate it into the SitAdapt observation environment. This research might also have some ethical consequences. Therefore, we follow the rules of value sensitive design [14] in our current and future activities.

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